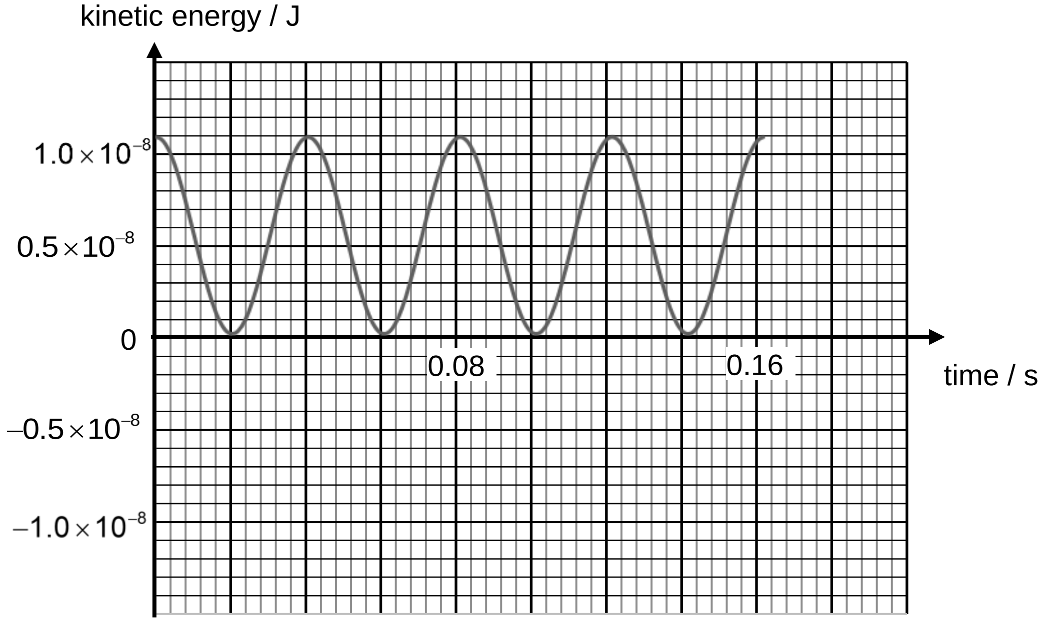
	Decay constant, $\lambda = \frac{\quad}{t_{1/2}} = \frac{\quad}{5.7 \times 10^3} = 1.22 \times 10$ year	
8(a)	Simple harmonic motion occurs when the acceleration of the object is directly proportional to its displacement from its equilibrium position and the acceleration is towards the equilibrium position/opposite to its displacement.	[1] [1]

8(b)(i)	$\omega = 2\pi f = 2\pi(33) = 66\pi$ $x = 2.1\times10^{-3}\sin\left[(66\pi)t\right]$	[1] correct amplitude and ω [1] any correct equation
8(b)(ii)	A distance of 0.8 mm to the right means the brush head is 1.3 mm from its equilibrium point. $x = 2.1\times10^{-3}\sin\left[(66\pi)t\right]$ $1.3\times10^{-3} = 2.1\times10^{-3}\sin\left[(66\pi)t\right]$ $t = 3.2\times10^{-3}$ s $v = \left(2.1\times10^{-3}\right)(66\pi)\cos\left[(66\pi)t\right]$ $= \left(2.1\times10^{-3}\right)(66\pi)\cos\left[(66\pi)\left(3.2\times10^{-3}\right)\right]$ $= 0.34$ m s^{-1}	[1] correct x [1] correct substitution [1] correct ans
8(c)	$v = 9.2\times10^{-2}\cos 77t$ $v_o = \omega x_o = 9.2\times10^{-2}$ $(77)x_o = 9.2\times10^{-2}$ $x_o = 1.2\times10^{-3}$ m	[1] correct v_o substitution [1] correct answer
8(d)(i)	$v = 9.2\times10^{-2}\cos 77t$ $\max K.E. = \frac{1}{2}mv_o^2 = \frac{1}{2}\left(2.5\times10^{-6}\right)\left(9.2\times10^{-2}\right)^2 = 1.1\times10^{-8}$ J	[1] correct K.E. substitution [1] correct answer
8(d)(ii)	 [1] for correct shape over 2 periods [1] for correct calculation of period T [1] for correct labelling of both axes	

	$\omega = \frac{2\pi}{T} = 77$ $T = 0.082$ s	
8(d)(iii)	Frequency is $f = \frac{77}{2\pi}$ Hz . The time interval is $\frac{3}{4}T = \frac{3}{4f} = \frac{3(2\pi)}{4(77)} = 0.061$ s.	[1] for $\frac{3}{4}T$ [1] ans
8(e)(i)1.	A forced oscillation is one which is driven <u>by an external force</u> such that energy is supplied to the oscillation.	[1]
8(e)(i)2.	The effect illustrated is called resonance.	[1]
8(e)(ii)	<u>same starting point</u> and <u>lower graph peak</u> maximum amplitude at same / lower frequency within original shape	[1] [1]
9(a)(i)	At constant pressure,	[1] for correct V_2